

Image Processing Based Dust Detection and prediction of Power using ANN in PV systems

Dania Saquib, Mohammed Nabeel Nasser
 Department of Solar and Alternate Energy
 Amity University, Dubai Campus
 Dubai, UAE

daniaasaquib@gmail.com, mohammednabeel1707@gmail.com

Swaroop Ramaswamy
 Department of Electrical and Electronics
 Amity University, Dubai Campus
 Dubai, UAE
 spillai@amityuniversity.ae

Abstract: -- Currently in the market, the most effective solar panels constitute the efficiency ratings as high as 22.8%, while majority of the panel efficiencies vary from 15% to 17%. However, the theoretical photovoltaic conversion efficiency reaches 86.6% [1]. This is mainly due to the fact that, it is assumed that each photon is optimally used and have high concentration ratio which is not the case for terrestrial solar panels and hence have higher efficiencies. On the other hand, different factors disturb the power output of a terrestrial module in which most of the factors are related to the module itself. In the UAE region, one of the key issues that affect the performance of the photovoltaic module is dust. This remains as a challenging task and many research works are also carried out. In this paper, image processing is used to detect the dust and the percentage of dust is used as a parameter to put into the neural networks. Irradiance depending on the location is taken as one input and the percentage of dust sensor is used to predict the voltage in the panel. If the voltage of the panel is reduced to certain limit then the signal can be triggered to clean the solar panel.

Keywords—Image processing, solar power forecasting, dust sensor

I. INTRODUCTION

R. Venkateswari et al [2] gives a detailed assessment of the issues that limit the power efficiency of a solar PV system. It is noted that, the materials showcases a vital part in the solar cell manufacturing from the core. It is found that, silicon is commonly used due to their plentiful availability and reduced cost. However, the efficiency of silicon photovoltaic module is low. Also, D.C. Jordan et al [3] reviews in paper about the degradation rates of photovoltaic module and it was found that solar panels usually degrade with an average rate of 0.5%/year and silicon based crystalline photovoltaic panels that degrade slower than thin film panels. One of the major factors which impacts the performance of a photovoltaic module is dust accumulation that remains as a major concern especially since the United Arab Emirates has a desert climate. UAE experiences a fair amount of dust storms throughout the year. The accumulation of high amount of dust settlements on the

photovoltaic module is mostly due to such dust storms that make it a critical element for the whole photovoltaic system [4].

UAE is categorized not only by dust particles but also by high temperature and humidity which are also the major factors which can also affect the efficiency of photovoltaic systems. Fig. 2 depicts the I-V features of a solar PV module under varying temperature. The most affected parameter in a solar cell when temperature increases is the open Voc. So, with an increase in temperature, Voc decreases. Therefore, the photovoltaic technology which operates within the desert climate as that of UAE requires more attention towards the operation and maintenance part of the system.

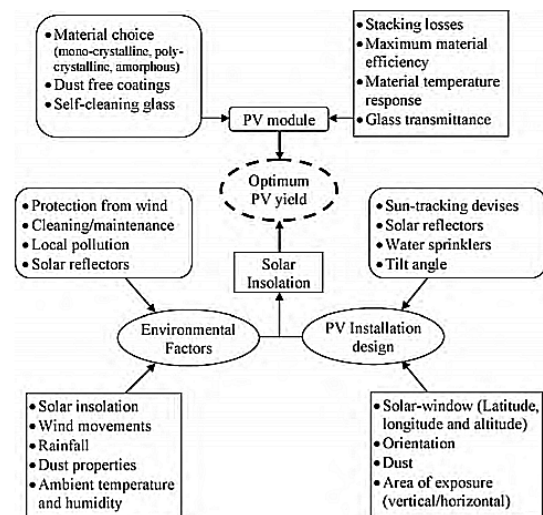


Fig. 1 – Factors that affect the performance

K.V. Vidyanandan [5] talks about the impact of soiling on photovoltaic module in his paper. Soiling is a label for used to define accumulation of dust and other pollutants on the photovoltaic module which tends to form a thin coating on the module and prevents large amount of light falling onto the

module. Dust is a small compact substance less than 500 μ m in diameter. The settlement of dust depends on factors such as dust properties, weather conditions, location, tilt angle of the panel and wind speed. Soiling can become permanent when humidity condensates and makes dust stick to the surface particularly at the bottom of the solar PV module. Dust deposition on the frame of the module can cause shading on the solar cells and also damage module coating as well.

Soiling in a photovoltaic system can cause power loss up to 17% per annum, which can be greatly reduced by regular cleaning. Many methods are available for PV cleaning. The cleaning frequency will differ depending up on the location, weather and type of mounting of the modules. Soiling is a significant factor in rising the operation and maintenance costs of photovoltaic plants.

II. IMPACT OF DUST

A.A. Hachicha et al. [6] reviews the influence of dust on the power output efficiency of solar PV system under UAE climate environments. They found that dust particles were small in size in which many of the particles were below 25mm having a greater impact on the efficiency as well as the electrical characteristics of the photovoltaic system. During high dust density, I_{SC} gradually reduced. The deposition of the dust depends on the orientation and tilt angle of the photovoltaic panel. Removal of dust is promoted due to the influence of gravity i.e. with increase in the tilt angle of the PV surface, dust deposition increases. Rainfall is random in UAE but it can help increase the photovoltaic efficiency and reduce the density of the accumulated dust. The operation and maintenance of PV modules cannot depend on infrequent rainfall and should be scheduled frequently depending on the accumulated dust density.

III. DUST CLEANING TECHNIQUES

Solar Photovoltaic modules have always been exposed to various types of climatic conditions and have high chances of developing dust, bird droppings, industrial residue etc. Some of the cleaning methods involve agents and chemicals which attract dirt and accelerate deterioration process with the consequence that effect the appearance and mainly the function of the solar panels. Due to the presence of these foreign material on the solar module it prevents the solar irradiance from entering the photovoltaic cells of the panel, therefore reducing the performance and efficiency [7]. It was found that accumulation of dust depends on various factors such as type of coating over the PV module, orientation, slope and surface roughness. Influence of dust particle accumulation depends on different properties of dust such as size, shape, weight and chemical properties of dust [8]. Saidan et al. [9] stated that regular cleaning of the solar panels is very important, else the magnitude of dust impact will be very high and there will be a huge difference between the actual yield and estimated yield of the power systems. Having the right cleaning mechanism is important to maintain the overall power output. Mani et al. [10] assessed the effect of dust over solar panels and observed that the quantity of dust that is accumulated over the surface effects

overall delivery from the module. The factors that affect the characterization of soiling accumulation on panels are (i) local environment property of dust (ii) property of dust. Patches of soil may include bird droppings, leaves block some cells in the panel, which severely affects the solar panel. There are two types of shading on the photovoltaic modules (i) Hard shading –This occurs when there are a number of solar cells that are shaded. If some cells are shaded and until and unless the unshaded cells receive solar irradiance, there will be a decrease in voltage of the photovoltaic module. (ii) Soft shading- this occurs during different environmental conditions such as smog.

A comparative study was done by Hussain et al. [11] on seven samples of dust on the photovoltaic modules, under three different radiations (850,750,650 W/m^2). The results stated that most of the solar irradiation has been blocked by the smallest particles.

IV. POWER FORECASTING MODELS

J. Antonanzas et al. [12] discusses about the different methods used to forecast solar power. The first method known as “white box” involves analytical equations to formulate the model of a photovoltaic system. The main focus to obtain this model is kept on obtaining accurate irradiance forecasts, which is one of the vital factors that affect the power production. In contrast to this the second option help directly predict the output power using machine learning and statistical methods. Some forecasts can also be made by mixing the above two methods, this method is known as “grey box” or hybrid model. Brabec et al. [13], has discussed the advantages and drawbacks related to this approach. The PV performance statistical method has more advantages when compared to the statistical method as previous data is not required, and it's also possible to predict the power out of the plant without the plant being constructed. Power output of the plant can be stated by knowing the net present worth (NPW) and the technical specifications. This lacks temporal and spatial resolutions with is one of the major cause of errors.

Lorenz et al. [14], found a new way to get rid of the errors by the Model Output Statistics (MOS) which can be used for weather forecasts. This method is useful as it improves resolution by interpolating values. Another method that could be used is by in cooperating the zenith angle with the clear sky index. Verzijlbergh et al. [15] have discussed a method using stepwise linear regression so as to obtain variables that best represent the error.

A. Statistical Model

These types of models don't require internal information to be transferred between the system and model. This method involves a data-driven approach that extracts the relations depending on the past data (past data may include meteorological and power datasets) to forecast the upcoming behavioral pattern of the power plant. More focus is kept on obtaining the historical data for accurate predictions. Gradioti et al. [16], stated that these models have been proven to be more superior to the PV performance models. The key benefit of this

model is the fact that it benefits from the ability of modifying systematic errors related to the inputs.

Almeida et al. [17] has associated the training sets according to the previous samples, on similar clearness index and resemblance between the empirical distribution on the irradiance prediction for day prediction and for each day of the dataset. They arrived at a conclusion that the best combination of the data sets was the adoption of 30 days, lesser to which result would be inadequate and more days would not give accurate outcomes. All this was based on the empirical distribution.

Mellit et al [18], Shi et al. [19] have developed models with promising results by dividing the training set based of the weather conditions (i.e. sunny ,cloudy and overcast), this data was then trained every day to be forecasted with a proper dataset.

B. Regressive methods

This method involves techniques that estimate relationships between independent variables called predictors and dependent variables such as solar power output. Considering the time series output weather stationary/non stationary and linear/nonlinear the classification is done. The difference between stationary and non-stationary is that stationary time series dedicate time series that vary around a static mean and non-stationary series don't involve mean.

C. Linear Stationary Models, Linear non-stationary models & Non-linear stationary models

Bacher et al. [20] has discussed about the Auto-Regressive models (AR), which formulate the output as a linear combination of the predictors and the lagged values. Li et al [21] on the other hand used simple moving average models, where data was present as constant variance over an equilibrium position around the mean, the average of this past data was used for the prediction. Double moving average models were to be considered when the data follows a certain trend.

Pedro et al. [22] worked on the Linear non-stationary models that have Auto-Regressive Integrated Moving Average models (ARMA). These models consider the union of Moving Average and Auto-Regressive models. Non – Linear stationary models followed the Nonlinear AR-exogeneous (NARMAX) model.

D. Artificial Neural Networks (ANNs)

J. Antonanzas et al. [12] has discussed about Artificial Neural networks which is a machine learning technique to forecast solar power. They work on neuron operations; many neurons are connected to form a neural network. Connections are made to each neuron and each connection has a certain weight, the final value of these weights are given by a training phase, combination of all this results in the predicted output. This method is useful when there are huge number of inputs. There are many different topologies, classification is done by n number of hidden layers depending on simple perception or multi perception.

V. POWER FORECASTING

Artificial neural networks (ANNs) have many different types of functions, namely prediction, regression and curve fitting. In our project, ANN has been used to formulate and predict the power output of the photovoltaic module. The most important part of an artificial Neural Network is the neurons that transfer functions which help formulate the output. Every input neuron present in the ANN is multiplied with weights which serves as associations between the input and neuron and this association is also between numerous layers of neurons. The last step involves the application of a transfer function to get the results. One of the main benefits of using ANN is that they don't require mathematical calculation knowledge among the input parameters but involve comparatively less computational effort and are able to generate. An Artificial Neural Network (ANN) having 0 hidden layers is known as perceptron, and the basic architecture is shown below. An Artificial Neural Network with one or more hidden layers is known as Multilayer perceptron. The number of hidden layers and the nodes present in each layer may vary depending upon the output to be obtain, usually the number of nodes required in each hidden layer can be calculated by simple techniques such as having a midway value between input and output , less than 2 times the input node , $2/3$ input nodes + output node.

VI. BACKWARD PROPAGATION

Backpropagation is also known as backward propagation of errors, which is basically a standard method of training ANN. It helps calculate a gradient of a loss function in terms of all the weights present in the network. It involves fine tuning of the weights in ANN depending on the rate of error obtained in the previous iteration (i.e. epoch). This is done in order to decrease the error rates and make the model accurate as it increases its generalization. Types of Backpropagation include:

A. Static Backpropagation

This type of backpropagation network produces a mapping for the static output from the static input. It is helpful in solving static classification problems such as optical character recognition.

B. Recurrent Backpropagation:

This type of backpropagation networks is fed forward until a fixed value is obtained. After which the error is calculated and is propagated backwards.

C. Forward propagation:

In order to create the propagations output activations, forward propagation is done on the inputs through neural networks.

D. Backward propagation:

Back propagation of the propagations output activations through ANN using training pattern target in order to generate the deference of all hidden and output neurons.

E. Weight Update

To obtain the gradient of the weight, the output delta and input activation are multiplied. A ratio of the gradient must be subtracted from the weight. The ratio mentioned above is a major factor that effects the quality and the speed of learning. Lower the ratio, accurate is the training and higher the ratio faster the neurons train. Phases 1 and 2 are repeated until satisfactory network performance is obtained.

VII. IMAGE PROCESSING

In image processing consists of three basic steps which includes importing the image. Manipulating the image and correct the image based on the analysis. In this the image is converted in to the function $f(x,y)$ where x and y are the coordinates and the f is the function based on the intensity of the image. The images can be classified as binary which consists of all the data between 0 and 1. 8 bit and 16 bit color image where there are combination of 256 and 65536 color respectively. There are various phases of image processing. The figure 2 below indicates the accumulation of dust in the panel. This picture is fed to the image processing tool to get the accumulation of dust. The image processing tool is used to predict the percentage of dust in the panel. IT can be seen that the percentage of dust is more than 70%. This data is fed to the neural network model to further predict the output.



Fig 2: Dust in the panel

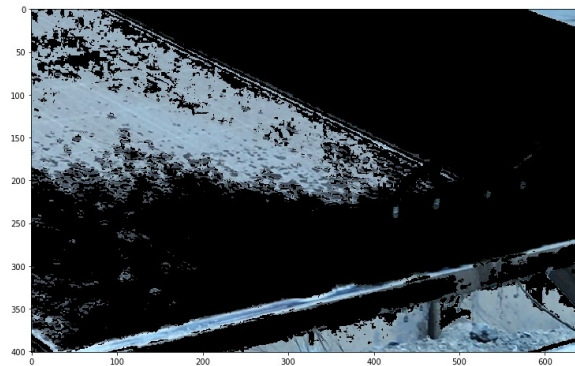


Fig 3: Image processing

VIII. DATA COLLECTION

The data has been collected with the inputs as dust and irradiance on the solar panels and the output as voltage and current. All the parameters such as irradiance, voltage and current have been measured using light dependent resistor, voltmeter and ammeter. The percentage of dust over the solar panel is calculated using image processing technique.

These two parameters (amount of dust and solar radiation) have been considered as inputs as they greatly affect the performance of the PV system. Dust has a negative impact on the output performance of the system thereby reducing the amount of radiation falling over the solar panel.

Following are the specifications of the 20W panel used :

Open Circuit Voltage (Voc)	-	21.2V
Peak Voltage (Vmp)	-	17.25V
Short circuit current (Isc)	-	1.27A
Peak Current	-	1.15A

Table below shows the database created for a 20W solar panel while varying the dust and light intensity we obtain different voltages.

DUST (%)	LDR (lx)	VOLTAGE(V)
1.6815	523	10.38
4.1121	517	10.29
7.657	517	10.16
11.1633	518	9.98
16.2977	515	9.83
20.0669	514	9.68
24.5581	518	9.48
29.0831	513	9.33
29.9196	514	9.19
36.0139	516	8.96
37.5699	509	8.88
40.3252	508	8.8
41.005	508	8.62
42.3834	507	8.46
48.0177	501	8.27
49.7133	500	8.08
51.2454	500	7.76
53.1713	496	7.54
57.3638	493	7.32
58.8671	494	7.18
61.1857	493	6.83
63.7854	491	6.42
68.9773	487	6.05
71.1491	491	5.69
74.3002	490	5.35

77.8157	480	4.05
78.1004	483	3.98
80.2217	468	3.65
80.5273	461	3.38
79.9116	461	3.3
82.4579	460	3.14
84.7826	453	2.95
86.7998	455	2.59
87.6457	448	2.25

IX. ARTIFICIAL NEURAL NETWORK IMPLEMENTATION

The design of this ANN model is important for the prediction of the power output (i.e. voltage) of the PV module based on the dust accumulated and the availability of solar radiation. Artificial Neural Network (ANN) has been created to contain 3 layers, they are input layer, output layer and hidden layer. The layer which receives the collected data is known as the input layer and the layer which produces the computed information is known as the output layer, one hidden layer which is helpful in connecting the output and input using processing units known as neurons. In an ANN model, the output value can be predicted by training input and output data sets. Here forward and backward propagation has been used. This ANN network has been developed in python, based on three layers using feed forward and backpropagation network with sigmoid function in the hidden layer. The subsequent parameters such as dust intensity, and irradiance are used as inputs and the outputs were the voltage and current.

X. BACK PROPAGATION ALGORITHM

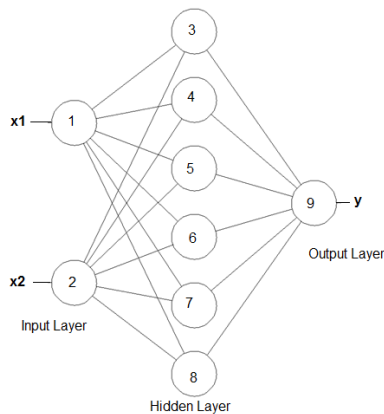


Fig 4: Neural Network model

The equation 1 shows the error in the model

$$\text{Network Error} = \text{Pred} - \text{Req} = E \quad (1)$$

The change in the weight is the subtracting old weight with the product of learning rate and the rate of change of error.

$$W_{AB(\text{new})} = W_{AB(\text{old})} - \eta \frac{\partial E^2}{\partial W_{AB}} \quad (2)$$

$$\frac{\partial E^2}{\partial W_{AB}} \quad (3)$$

is the sensitivity error,

From the chain rule,

$$\frac{\partial E^2}{\partial W_{AB}} = \frac{\partial E^2}{\partial I_B} - \frac{\partial I_B}{\partial W_{AB}} = O_A \quad (4)$$

So the new weight can be rewritten as

$$W_{AB(\text{new})} = W_{AB(\text{old})} - \eta \frac{\partial E^2}{\partial I_B} O_A \quad (5)$$

The equation 5 indicates the final weight change which is obtained by subtracting old weight by the product of learning rate, rate of change of error with input and the output.

In figure 4 it can be seen that dust and the irradiance are given as the two inputs. Here one hidden layer, one input layer and one output layer is used. In the hidden layer 9 neurons are used. The trial is run for various possibilities by changing the number of neurons in the hidden layer and by changing the number of iterations. It can be seen that after 150000 iterations the model shows less error. The loss between the actual and predicted power accounts to about 0.000146, which is very minimal. Figure 4 as seen below shows the difference between the actual and predicted voltage output.

The figure 5 indicates the error curve and it can be seen that the error keeps reducing with the increase in number of iterations.

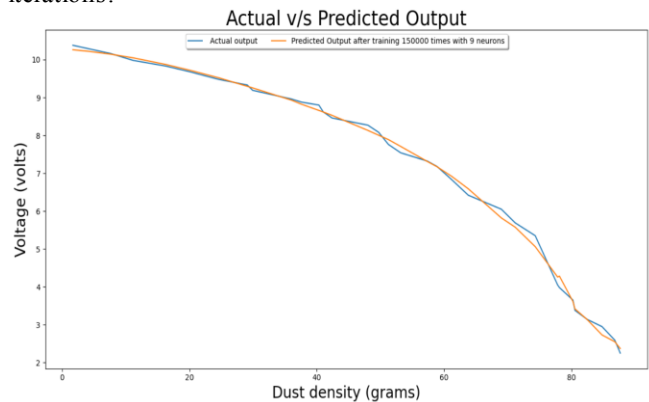


Fig 4: Actual and the predicted output

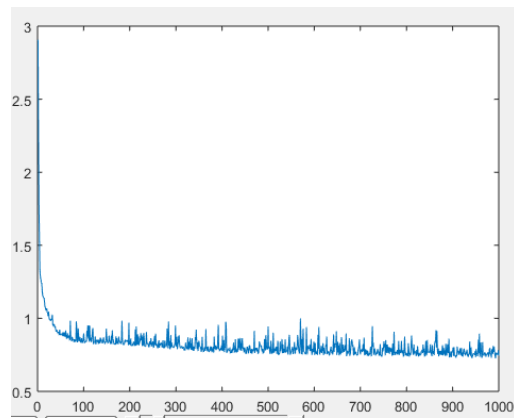


Fig 5: Error curve

XI. RESULT AND DISCUSSION

Figure 6 show the presence of dust over the 20W panel before and after detecting respectively and the detection of dust has been done by image processing.

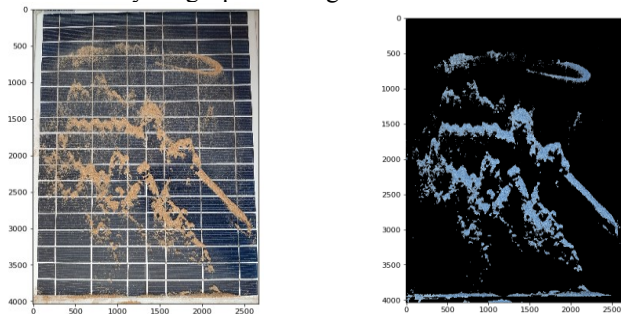


Fig 6: Before detection and After detection

The result obtained from the algorithm developed is as follows:

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Phase 1
Dust Percentage : 13.357245561390346 %
Phase 2
Predicted Voltage :9.4681792 V
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Fig 7: Depiction of dust percentage and voltage prediction

When this is compared to the actual voltage which is 9.47 V. The error accounts to about 0.01 V. This implies that the model created for power forecasting is able to predict values very close to the actual output.

XII. CONCLUSION

In this paper image processing is used to predict the dust. It can be seen that it is possible to detect the percentage of dust in the panel. This data is used along with the irradiance to predict the output. Though it is demonstrated for a single panel, it can be used to predict the number of panels for a particular area by multiplying the number of panels used, as the dust intensity and the irradiance will be same for the

particular region. This can be further extended to the number of inputs by adding other parameters like temperature, humidity which affect the performance of the panel

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